Resume

Rupak Roy

Portfolio Link: https://www.linkedin.com/in/bobrupak/

In brief:

Over the years of experience with my motto to seek the best of skills and opportunity for a data-driven scientific approach & framework in a progressive organization have made me competitive in the Analytical era of Information Technology. With my diverse experience spanning roles as an individual contributor, solution architect, team leader, project manager, and delivery manager. I bring a well-rounded view to delivering end-to-end data science solutions. This versatility allows me to drive impactful, outcome-focused results across projects.

Currently associated with Genpact designated as Sr. Manager for Data Science Advanced Analytics, GenAi, MLops as well the LLMOps to integrate best practices at a scalable, production-ready product development..

Forward-thinking and enterprise problem-solver with tactical foresight and proven success capitalizing on growth opportunities, improving bottom-line performance and optimizing efficiency, productivity and profitability through delivery of value-added services and procedures.

Articulate Communicator and effective trainer with a direct and decisive management style that focuses on matrix teamwork and clearly defined goals and values for buy-in of all levels.

An Out-of-the-box thinker with a proven track record of increasing revenues, evolving process and productivity improvement with an innate ability to work in unison with customers as well as the staff.

Employment Scan / Career Contour

2024 -

Company: Genpact
Designation: Senior Manager GenAi & Data Science
Roles and Liabilities:
* Lead and mentored a team of Junior to Senior Data Scientists as well Data Science Managers, ensuring seamless communication and integration of best-in-class data science and GenAI solutions.
* Lead project delivery while designing and implementing end-to-end architectural solutions for Data Science & GenAi pipelines and applications, including budget, resource estimation, infra estimation, and delivery planning.
* Served as a Solution Architect, responsible for developing scalable, efficient, and innovative Gen AI & DS driven solutions aligned with business needs.
* Develop & fine-tuning Gen AI models such as GPT for various applications.
* Consulted and demonstrated POCs to Points of Sale (POS), showcasing data-science and GenAI capabilities.
* Evaluated GenAI model performance and implemented improvements based on feedback.
* Ensured ethical AI practices, addressing bias, privacy concerns, and fairness in AI models.
* Partnered with data scientists, engineers, and functional SMEs to integrate Gen AI into products & services for automation, predictive modeling, synthatic data generation, and actionable insights.
* Overseeing the complete SOW for the GenAi LLMOps lifecycle, including the evaluation of Gen AI model performance across L1, L2 and L3 layers, ensuring operational efficiency, reliability, and continuous improvement.

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Featured on
Medium @bobrupakroy
Udemy @ai-master-class
pypi.org

Core Competencies

- Data Science
- ML Ops/LLMOPS
- Generative Ai
- Azure/AWS
- Agile Scrum Master
- PMP
  - RMP Risk Management
  - PgMP Program Management
- Six Sigma Green Belt
- Lean Management, PMO
- Technology Transfer.

Employment Scan / Career Contour

2020 - 2024

Company: Amdocs
Designation: Lead Data Scientist: Advanced Analytics
Seniority : Advanced
Roles and Liabilities:
* Data mining and develops algorithms that provide insight from data.
* Developed metrics for assessing accuracy and generalization of the trained models.
* Assigned to/for data science oriented value proposition via ml marketing campaigns, product development and automation.

* Closing working with solution architect, pre-sales lead as well reporting to Head to Engineering for consulting & demonstrating the POCs to POS.
* Leading a team of operations to communicate and integrate the next gen. hybrid/ genetic based predictive models, mlops as well GenAi to the existing engine.
* Recruited data scientist, budget, roadmap, milestones & the logistics combing multi-role skillsets.

Success Story/White-paper:
* https://www.amdocs.com/insights/article/using-ml-drive-postpaid-migrations
* ADI Innovators Award 2024 for Data Science

2014 - 2020

Company: Kommlabs Pvt. Ltd.
Roles and Liabilities as Sr. Engineer/Data Science (Team Size: 4)
* Handling 4 Projects with team size 4 combining 18servers from Nas to Central Database Server, Mediation Server as well as Store & Forward Server solely responsible for SDLC of the data mgnt. projects and the Analytics as well as troubleshooting and training the client.
* Training all new colleagues in the insights and logistics of the organization including big data management and the analytics for the company procedures.

Key Highlights :

* Certificate of Appreciation from Amdocs 2021, 22
Appreciation Certificate for excellent performance from State Government. 2014,15,16,18

Some Of The DS/AI-ML SDLC Project Insights:

Project Catalog Link: Data Science/Analytics

Best Offer (Generative Ai): Streerable LLM with multi-agent LLM chain showcasing user authtication using llm for accessing part LLM agents or full LLM agent capabilities, pro-active offer prediction chain (NBO) by integrating ml models, Rag Fusion chain, Query Chain, session memory for Proximal Policy Optimization(PPO) for PEFT with 2- layer sematic routing firewall to avoid LLM hallucination for the Banking BFSI space.

Report Consolidation (Generative Ai): Developed a business llm with the capability to consolidate multiple sql queries taking into consideration important measures, attributes & predicates as well as understanding data model, relation and more.

Infra Search Engine (Generative Ai): Implemented multi-approach search using crawler for faster index search with Rag Fusion, Corrective Rag as well Canopy Rag techniques in Japanese with a goal to increase the quality of the search results.

Order Management LLMOPS L2 Support (Generative Ai):
The GenAI-based solution extracts structured information from unstructured email inputs. It streamlines sales order creation by feeding the extracted data into a bot for SAP processing. Develop a clear understanding of the existing architecture, supporting documents, and LLMOPS SOW.Define the project scope, allowing minor enhancements (up to 4 hours) and RCA activities, while excluding major upgrades like new model integrations or structural changes. Establish clear communication channels, actively support hypercare, and be fully prepared for different outage scenarios with defined roles and responsibilities

Analytical Tools

Python, R & Knime

Microsoft Sql Server

Tableau/Power Bi

SnowFlake

Hue/Hive

Others

Sql, Scoop, Flume, MapReduce, Hive, Linux, Power BI

Advance Analytical Abilities

- Custom Ensemble modeling

-- Voting, Stacking, Binding

- Hybrid Models
- Outlier Detection Models

-- Isolation Forest, MCD, LOF,

-- OneClassSVM, DBSCAN.

- MARS - Multivariate Adaptive Regression Splines
- Synthetic Sample Generation

-- SMOTE-ENN, SMOTE- TOMEK

-- PAR- Probabilistic AutoRegressive

-- Triplet based Variable AutoEncoders (TAVE)

-- CopulaGAN

- Classifier and Regressor Chain
- Huber, RANSAC, Theil Sen Regression

Analytical Research

Publisher: Issuu

Category:
- Face Recognition
- Data flow and ML applications in:
- medical & healthcare, financial data, marketing & advertising, food/culinary industry, law & enforcement, census, fashion, manufacturing & production.

Industry Solutions Deployed at Production Level (Azue/Aws)

Asset Depreciation for tax forecasting: Managing large asset portfolios is complex due to differing financial and regulatory classifications. Annual asset value updates are critical, as errors in depreciation can impact income taxes and risk surcharges or fines if identified late.

Invoice Exception: The retailer faces 6 million annual invoice exceptions, requiring 750 employees for resolution due to mismatched IDs, SKUs, prices, and quantities. Current matching is limited to summary-level rules without advanced capabilities. The process uses metrics like Levenshtein distance and Jaro-Winkler similarity to match invoices and receivings, quantifying mismatches and providing recommendations based on match thresholds.

Prioritizing Invoices: Invoices face roadblocks, causing delays or being sent back to previous teams. Productivity issues among agents and teams impact efficiency, while the absence of an optimal prioritization process and incorrect routing contribute to further delays.

Bad Debt Prediction: The account is experiencing rising bad debt and pressure from the Sales Team to relax payment terms. To address this, there is a critical need for better risk management and credit decision-making, prompting the search for a predictive solution to assess credit risk. The goal is to build a prediction model to score customers based on their risk of becoming bad debt, enabling proactive actions like credit limit adjustments, credit blocks, or payment term changes.

Treasury Cash Flow Forecasting: Inefficient short-term cash forecasting leads to excess cash holdings, while manual processes and outdated systems delay real-time insights. Key forecasting data includes AR/AP invoices, payment due dates, transaction details, and historical cash flow. The goal is to reduce excess cash holdings by 30% (from €1 billion to €700 million), improve cash management, and save on interest costs with better forecasting.

Predictive POT: A POT of 91% is straining supplier experience (73% of vendors) and weakening procurement negotiation power. Delayed payments and inaccurate reporting lead to penalties, adverse publicity, and potential business sanctions. Predictive analytics provide early warnings, identify root causes, and offer scalable solutions for remediation across regions.

Enhancing Oncology Analogue Library with Commercial KPIs - Sensitivity Analysis: The business use case involves understanding oncology indications, competitor landscape, and product profiles to estimate market share impact and develop "Time to Peak" and "Peak Market Share" for sensitivity analysis. A regression model was developed to forecast these metrics, categorizing market share uptake as "Fast/Medium/Slow" with a simulator for scenario comparison and predictions.

Smart Media Monitoring Plus (Generative Ai): The team creates a daily report summarizing media coverage of corporate news, brand mentions, and competitors in the Latin region. The solution monitors multiple online sources, offers advanced search filters, relevancy checks, and ensures content is not from homepages or PDFs. It features automated reporting, structured data collection, and strong admin capabilities for efficient historical analysis.

Ai Knowledge Assistant Tool Migration(GCP)/LLMOPS (Generative Ai) : The business needs to migrate the tool from the dev to the prod environment, with LLMOPS support for 24/7 operations. The approach involves understanding the current architecture, reviewing supporting documentation, and defining scope, including minor enhancements (up to 10 hours) and exclusions like new LLM models, structural changes, and model fine-tuning. analysis.

Driving Mail Trafic (Generative Ai): The solution automates email identification, extracting order details and sending auto-responses to the relevant recipients. It uses conversation history and Reinforcement Learning from Human Feedback (RLHF) to enhance contact identification, updating the LLM route prompt list for better communication. The approach leverages RLHF without PEFT, ensuring cost-effectiveness.

PriceFlex MicroBuild: The retail industry faces challenges in balancing competitiveness and margin protection due to limited competitive visibility, complex pricing models, and poor collaboration with pricing teams. Retailers need an intelligent, automated pricing solution to make data-driven decisions. The solution integrates multi-model analytics to identify sales trends, perform sensitivity analysis, and provide rule-based insights like bundling recommendations and margin detection. An interactive GUI allows deeper exploration, helping managers set optimal pricing strategies and understand demand patterns.

Advance Analytical Abilities

- CNN:
  - -- Sentiment Analysis
  - -- Multivariate Multiple Input Series.
  - -- Multivariate Multi Headed Series
  - -- Multivariate Parallel Input Series.
  - -- Multi-Step Input Series and more.
- LSTM:
  - -- Multivariate TS Forecasting.
  - -- Multi-Step TS Forecasting.
  - -- LSTM Regression Analysis
- Hybrid CNN-LSTM-MLP
- Deep Learning-Grid Search
- AutoEncoders, T- SNE Anomaly/Fraud Detection
- Fuzzy, BRICH, DBSCAN, Optics
- Probability Caliberation
- Bregman Divergence Selector
- Bias-Variance Decomposition
- and more.

Web Framework :

- Streamlit Python

Cloud Framework :

- GCP, Azure
- AWS SageMaker
- Heroku:
  @ streamlit-roy.herokuapp.com

Leadership Certifications :

- PMP & Agile Scurm Master.
- Lean Management & Six Sigma Green Belt
- PMI-RMP - Risk Management
- PMI-PgMP - Program Management and PMO

Industry Solutions Deployed at Production Level (Azue/Aws)

System Health: Automated the process of predicting and monitoring system resources with feedback loops for immediate re-training accross the big data jobs to have synchronized fail-safe resource allocation for marketing campaigns.

Identify and Automated Personalized targeting to meet the individual needs.
Applied state of the art hybrid deep learning forecasting models ensembled with classical ml models to sucessfuly identify the active patterns daily, weekly and overall trend without compromising the deep packet inspection protocol.

Application Error Detection and Forecast
Live Anomaly detection accross operations to give management fine decision making information for O&Ms as well with downtime forecasted dates with RCA

Ai Studio: Developed a Gui based auto-ml that includes production-level data preprocessing to model building that detects data pattern and triggers the respective ml learning stack of supervised, unsupervised, timeseries & ensemble with auto deployment and MLops.

Customer Migration:Sucessfully able to meet the business needs by reverse engineering churn concept with FN metrics contributing to 1.3m ROI in a row.
Success story / whitepaper:
https://www.amdocs.com/insights/article/using-ml-drive-postpaid-migrations

Reactivation: Sucessfully uplift reactivation behavior of 3% of the total base to 18% around 200% rise with probablistic target resampling and encoding.

Best Offer Model: Alternative to supervised model using Thomspon sampling with clustering(patented) that can be used when we dont have enough info for a subscriber

Best Top-up Amount: Simulated increase in max topup when average last 3 topup is greater than median of last 3 topup offer the next max topup-else retain the previous behavior.

Best Time To Comunicate: Simulated forecasting propensity scores with classifier models using consolidated KPI's to understand the best time to communicate into high-mid-low value subscriber for marketing campaigns.

Customer Discrepency model: Implimented end2end mlops pipe using azure mlflow with L1, L2 and L3 support process.

Usage/Service spend: Anomaly detection in the subscription review, if expired it goes to pay-as-you-go(PayG) with a target to keep them in track in bundle as well for proactive campaign.

Einstein Discovery answers what happened, why it happened, what could happen, what will happen and how can we improve it.

Computer Vision: Designed and directed computer vision solutions for law & enforecement, logistics, manufacturing, fraud detection as well other Medical Image Processing.
* Few more includes Emotion-Detection, Motion Heat Map, Environment Detection, Human activity recognition, and Pose Estimation.

Training: Data Science, Ai, Big Data across platforms like Amdocs, Udemy, as well inother institutions as well corporate clients about the insights and logistics of data science & Ai.

Analytics: Hypothesis Testing for accessing the project performance with various probability distributions, Paired two-fold cross-validation with 5 repeats, Wilcoxon Signed-Rank, Kolmogorov–Smirnov and etc.
* Conditional probability re-balancing for Control Group (UCG) to Target Group (UTG)
* Applied pythagoras theorem to autotomate elbow for Clustering to find the category of patterns for the product usuage along with Eclat modeling for consumer purchase behavior.

ML orchestration using knime analyics to overcome the challenges of KT and on-prem solution.

Education: Master of Science in Information Technology: Msc IT, Smu
Massachusetts Institute of Technology: MIT, MicroMasters
PGD in Master of Commerce / International Business Operations

Certifications: Data Science, Wiley Big Data, Fraud Analytics, Text Mining, Coursera ML Ops & Coursera Gen Ai, PMP, Lean Management & Six Sigma Green Belt

ML Utilities

- Auto Impute Missing values
- Auto Skewness correctin
- Auto Data-Drift Detection

Generative Ai

- Multi-shot prompt eng.
- Parameter Efficient FineTuning
- RLHF
- Proximal Policy Optimization
- Program Aided Language Models
- LangChain
- Self Corrective Rag
- Rag Fusion
- Canopy Rag
- GPT 3.5 turbo/instruct/4
- Llama 2/ Mistral

Pypi Contribution:

- Pip install ~
  -- auto-feat-selection.
  -- auto-ts-ensemble
  -- autoroc
  -- autoelbow
  -- unsupervised Feat. Importance

Web Framework :

- Streamlit Python

Cloud Framework :

- GCP, Azure
- AWS SageMaker
- Heroku:

Personal Details :

Zodiac: Leo
Hobbies: playing indoor games, watching movies, photography and keeping myself upto dated with new technologies.
Strengths

- Expertise in Troubleshooting.
- Worked independently as well as part of a team.